Bilal Nadeem Khan

B.E. Student — Artificial Intelligence & Machine Learning

+91 9008493253 | bkbilalnadeem@gmail.com | Bengaluru, India

linkedin.com/in/bilal-nadeem-khan-2504b127a | **Github**/Bilal61355

OBJECTIVE

Final-year B.E. student in AI & ML seeking a Machine Learning internship where I can contribute to real-world projects while deepening my skills in data preprocessing, model building, and deployment. Passionate about applying academic knowledge to solve practical problems through Python, scikit-learn, and deep learning frameworks.

EDUCATION

B.E. in Artificial Intelligence & Machine Learning *2021 – 2025*

AMC Engineering College, Bengaluru (VTU) | CGPA: 8.1 / 10

Relevant Coursework: Machine Learning, Deep Learning, Natural Language Processing, Data Structures & Algorithms, Statistics & Probability, Computer Vision, Big Data Analytics

ACADEMIC PROJECTS

Sentiment Analysis & Opinion Mining System *Jan 2025 – Mar 2025*

Tech: Python · NLTK · Scikit-learn · TF-IDF · Flask · Pandas

- Built a text classification pipeline to predict sentiment (positive/negative/neutral) from product reviews using TF-IDF vectorization and Logistic Regression.
- Applied NLP preprocessing — tokenization, stopword removal, and lemmatization — using NLTK to clean raw text before model input.
- Achieved 87% accuracy on a test split of 2,000 reviews and deployed the model via a simple Flask API for live predictions.
- Visualized results using Matplotlib confusion matrices and word cloud plots to communicate findings clearly.

Crop Disease Detection using CNN *Aug 2024 – Nov 2024*

Tech: Python · TensorFlow · Keras · OpenCV · Transfer Learning (VGG16) · Streamlit

- Trained a CNN image classifier on a publicly available plant disease dataset to identify 10 common crop diseases from leaf photographs.
- Used Transfer Learning with VGG16 pretrained on ImageNet; fine-tuned the last layers, achieving ~91% validation accuracy.
- Applied image augmentation (flipping, rotation, zoom) via Keras ImageDataGenerator to reduce overfitting on a limited dataset.
- Built a Streamlit interface for local demo — users upload a leaf image and receive a predicted disease label with confidence score.

Customer Churn Prediction with Explainable AI *Mar 2024 – May 2024*

Tech: Python · Scikit-learn · XGBoost · SHAP · Pandas · Seaborn · Jupyter Notebook

- Predicted customer churn on a telecom dataset using multiple classifiers — Logistic Regression, Random Forest, and XGBoost — and selected the best via cross-validation.
- Performed exploratory data analysis (EDA) and feature engineering to identify key churn indicators: contract type, tenure, and monthly charges.
- Applied SHAP values to generate feature-importance explanations, bridging the gap between model predictions and business interpretability.
- Tuned hyperparameters using GridSearchCV; final XGBoost model achieved AUC-ROC of 0.89 on the test set.

TECHNICAL SKILLS

Languages	Python, SQL, R (basics)
ML / DL	Scikit-learn, TensorFlow, Keras, XGBoost, OpenCV
NLP	NLTK, spaCy, Hugging Face Transformers, TF-IDF, Word2Vec

Data & Viz	Pandas, NumPy, Matplotlib, Seaborn
Tools	Git, GitHub, Jupyter Notebook, Google Colab, VS Code, Streamlit, Flask
Concepts	Supervised & Unsupervised Learning, Neural Networks, Transfer Learning, Explainable AI, NLP Pipelines, PCA

CERTIFICATIONS

Oracle Cloud Infrastructure (OCI) AI Foundations *Oracle University*
Covers OCI AI services, ML fundamentals, and cloud-native AI deployment.

Python for AI and Data Science *IBM — Coursera*
Covers Python, Pandas, NumPy, Matplotlib, and foundational data science workflows.

Currently pursuing: Machine Learning Specialization — Andrew Ng (Coursera)

ADDITIONAL INFORMATION

- Strong problem-solving mindset with ability to learn new tools and frameworks quickly.
- Comfortable working independently and in team settings; experience collaborating on academic group projects.
- Familiar with Agile workflows and version control via Git & GitHub.
- Passionate about staying current with ML research through blogs, papers, and online courses.

References available upon request